

Jake Brawer, Ph.D.

[website](#) / [github](#) / [email](#) / [linkedin](#)

RESEARCH OVERVIEW

Summary I build **interactive AI systems** that combine perception, reasoning, memory, and learning to support robust human-centered interaction. My work spans **robotics**, **conversational AI**, and **production ML systems**, with emphasis on realtime decision-making, structured knowledge, personalization, and deployable architectures.

PROFESSIONAL EXPERIENCE

Nov 2025-Current **Founding AI/ML Engineer**
Stealth startup (voice AI / digital health)
Sole architect and founding engineer for an early-stage phone-based conversational AI platform for older adults.
Built the end-to-end system across telephony, realtime speech, LLM orchestration, long-term memory, FastAPI services, PostgreSQL, and AWS infrastructure.
Implemented platform safeguards for reliability and HIPAA-oriented handling of sensitive user data.

2023-2025 **Postdoctoral Researcher**
University of Colorado, Boulder,
Advisors: Alessandro Roncone and Bradley Hayes
Lead research on human-robot teaming, assistive robotics, and AI decision support.
Develop models for inferring human proficiency, trust, and intent from gaze and behavior in collaborative environments.
Build semantic robot perception systems for robust localization in cluttered indoor workspaces using low-cost visual-inertial sensing.

Spring 2024 **Lecturer, Advanced Robotics** (CSCI 4302/5302)
University of Colorado, Boulder
Sole lecturer for a graduate-level robotics course covering motion planning, SLAM, optimization, and reinforcement learning.

EDUCATION

2016-2023 **Ph.D. in Computer Science** *Yale University*, Advisor: Brian Scassellati
2012-2016 **B.A. in Cognitive Science; Computer Science minor** *Vassar College*

SKILLS

Languages and Systems **Python, C/C++, C#, SQL, Git, Emacs, Docker, Linux, $\text{\LaTeX}$**

AI / ML	PyTorch, scikit-learn, SciPy, NumPy, Pandas, LangChain, LLM orchestration, RAG / vector retrieval, PGVector, matplotlib, NLTK, Gurobi, Prolog, PDDL
Backend	FastAPI, PostgreSQL, AWS, async Python, REST APIs, system design, deployment, load testing, observability, HIPAA-oriented system design, PHI-safe logging
Voice AI	Pipecat, Deepgram, AWS Bedrock, Telnyx, Cartesia
Robotics and CV	ROS, Gazebo, RViz, OpenCV, Open3D, Baxter, Sawyer, UR5, Kinect, RealSense, RGB-D cameras
Soft Skills	Clear technical communication, Collaborative, Fast learner, Leadership

JOURNAL ARTICLES

- Shivendra Agrawal, **Jake Brawer**, Ashutosh Naik, Alessandro Roncone, and Bradley Hayes (2026). “**ShelfAware: Real-Time Visual-Inertial Semantic Localization in Quasi-Static Environments with Low-Cost Sensors**”. In: *IEEE Robotics and Automation Letters*. Accepted.
- Meiying Qin, **Jake Brawer**, and Brian Scassellati (2022a). “**Robot Tool Use: A Survey**”. In: *Frontiers in Robotics and AI* 9, p. 369.
- * Meiying Qin, * **Jake Brawer**, and Brian Scassellati (2021). “**Rapidly Learning Generalizable and Robot-Agnostic Tool-Use Skills for a Wide Range of Tasks**”. In: *Frontiers in Robotics and AI* 8, p. 380.
- Jake Brawer**, Aaron Hill, Ken Livingston, Eric Aaron, Joshua Bongard, and John H Long Jr (2017). “**Epigenetic Operators and the Evolution of Physically Embodied Robots**”. In: *Frontiers in Robotics and AI* 4, p. 1.

CONFERENCE PROCEEDINGS

- Nikhil Hulle, Stéphane Aroca-Ouellette, Anthony J Ries, **Jake Brawer**, Katharina von der Wense, and Alessandro Roncone (2024). “**Eyes on the Game: Deciphering Implicit Human Signals to Infer Human Proficiency, Trust, and Intent**”. In: *33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN 2024)*. Accepted for publication, proceedings pending.
- Jake Brawer**, Debasmita Ghose, Kate Candon, Meiying Qin, Alessandro Roncone, Marynel Vazquez, and Brian Scassellati (2023). “**Interactive Policy Shaping for Human-Robot Collaboration with Transparent Matrix Overlays**”. In: *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*. IEEE.
- Meiying Qin, **Jake Brawer**, and Brian Scassellati (2022b). “**Task-Oriented Robot-to-Human Handovers in Collaborative Tool-Use Tasks**”. In: *2022 31st IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)*. IEEE.
- Jake Brawer**, Meiying Qin, and Brian Scassellati (2020). “**A causal approach to tool affordance learning**”. In: *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, pp. 8394–8399.
- Brian Scassellati, **Jake Brawer**, Katherine Tsui, Setareh Nasihati Gilani, Melissa Malzkuhn, Barbara Manini, Adam Stone, Geo Kartheiser, Arcangelo Merla, Ari Shapiro, et al. (2018). “**Teaching Language to Deaf Infants with a Robot and a Virtual Human**”. In: *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, p. 553.

Zong Xuan Tan, **Jake Brawer**, and Brian Scassellati (2018). “**That’s Mine! Learning Ownership Relations and Norms for Robots**”. In: *Thirty-second AAAI conference on artificial intelligence*.

WORKSHOP PROCEEDINGS

Ava Abderezaei, Chi-Hui Lin, Joseph Miceli, Naren Sivagnanadasan, Stéphane Aroca-Ouellette, **Jake Brawer**, and Alessandro Roncone (2025). “**Towards Zero-Shot Coordination between Teams of Agents: The N-XPlay Framework**”. In: *Workshop on Multi-Robot Systems. Robotics: Science and Systems (RSS)*. Workshop paper; proceedings forthcoming.

Jake Brawer, Kayleigh Bishop, Bradley Hayes, and Alessandro Roncone (2023). “**Towards A Natural Language Interface for Flexible Multi-Agent Task Assignment**”. In: *Proceedings of the AAAI Symposium Series*. Vol. 2. 1, pp. 167–171.

AWARDS AND FELLOWSHIPS

July 2023, 2024, 2025 **Army Educational Outreach Program (AEOP) Fellowship**, University of Colorado Boulder — awarded in 2023, 2024, and 2025 to support postdoctoral research.

March 2023 **Best paper (technical track)**: Our paper “Interactive Policy Shaping for Human-Robot Collaboration” won the best paper award in the technical track at the 2023 International Conference on Human-Robot Interaction (HRI 2023).

WORKSHOPS ORGANIZED

March 2023 **Human-Machine Teaming Paradigm Workshop**
Jake Brawer, Bradley Hayes, Alessandro Roncone

March 2018 **Bridging the Gap: An NSF Workshop on Conversational Agents and Human-Robot Interaction**
Justine Cassell, Brian Scassellati, Jake Brawer, Michael Madaio
NSF Cyber-Human Systems (CHS), Robust Intelligence, National Robotics Initiative. Award #1829237

ADDITIONAL TEACHING EXPERIENCE

Spring 2018/2019 **Intelligent Robotics**
Teaching Assistant

Fall 2017 **Object Oriented Programming**
Teaching Assistant

Fall 2014/2015 **Perception and Action**
Teaching Assistant

SERVICE

Leadership Area Chair, International Conference on Human-Agent Interaction (**HAI**; 2024)

Conference Reviews	RSS (2025 — Outstanding Reviewer Award), HRI (2017–2025 — Outstanding Review Mention 2023), IROS (2020), ICRA (2019), Humanoids (2018)
Journal Reviews	IJRR (2024), T-RO (2024), THRI (2019–2025)
Student Supervision	Shivendra Agrawal (2024–2025) Ava Abderezaei (2024–2025) Gyaning Kumar (2025) Yi-Shiuan Huang (2025) Chi-Hui Lin (2024–2025) Aaquib Tabrez (2023–2024) Kaleb Bishop (2017–2019, 2023) John Dallard (2021–2022) Sarah Widder (2017–2019) Kevin Choi (2018) Acshi Haggemiller (2016–2017) Tan Zong Xuan (2017–2018)